JDBC Steps

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Objectives

- Steps for using JDBC.

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JDBC Steps

- There are 5 steps to connect any java application with the database using JDBC.

- Register the Driver class
- Create connection
- Create statement
- Execute queries
- Close connection

Register the Driver Class

- The forName() method is used to register the driver class. This method is used to dynamically load the driver class.
- Syntax of forName() method

```
public static void forName(String className)
    throws ClassNotFoundException
```

```
Class.forName("com.mysql.cj.jdbc.Driver");
```

Create the Connection Object

- The getConnection() method of DriverManager class is used to establish connection with the database.
- public static Connection getConnection(String url)throws SQLEx ception
  - public static Connection getConnection(String url,String name,String p assword)throws SQLException

Create the Connection Object

- Connection con = DriverManager.getConnection ("jdbc:mysql://localhost:1433/example" + "user=root&password=amma");
- Create an object of Connection and setup the connection with the DB

Create the Statement Object

- The **createStatement()** method of Connection interface is used to create statement. The **statement object is responsible to execute queries with the database.**
- public Statement createStatement()throws SQLException
- Statement stmt=con.createStatement();

Create the Statement Object

- Statement st = con.createStatement();

- The JDBCStatement, CallableStatement, and PreparedStatement interfaces define the methods that enable you to send SQL commands and receive data from your database.

Execute the Query

- The executeQuery() method of Statement interface is used to execute queries to the database. This method returns the object of ResultSet that can be used to get all the records of a table.

Syntax:

```
public ResultSet executeQuery(String sql)throws SQLException
```

```
ResultSet rs= st.executeQuery("select * from batch.student");
```

Close the Connection Object

- By closing connection object statement and ResultSet will be closed automatically. The **close()** method of Connection interface is used to close the connection.

Syntax:

```
public void close()throws SQLException

con.close();
```

Takeaways

- Steps for using JDBC

Reference

[1]. Java How to Program Dietel,Harvey Tenth Edition

[2].https://docs.oracle.com/javase/tutorial/jdbc/basics